

Activity-1 : Classes and Objects - Inheritance

Q1. Program based on Classes and Objects - Initializing Data member using Objects directly

Program:

```
import java.util.Scanner;

public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        Student s=new Student();
        s.rollNo=sc.nextInt();
        sc.nextLine();
        s.name=sc.nextLine();
        s.marks=sc.nextDouble();
        s.displayDetails();
    }
}

class Student{
    public int rollNo;
    public String name;
    public double marks;

    public void displayDetails(){
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+ marks);
    }
}
```

Input:

0

krish

54.50

Output:

Roll No: 0

Name: krish

Marks: 54.5

Q2. Classes and Objects - Initialize Object Data Using Setter Methods

Program:

```
import java.util.Scanner;

public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        Student s=new Student();
        long r=sc.nextLong();
        sc.nextLine();
        String n=sc.nextLine();
        double m=sc.nextDouble();

        s.setRollNo(r);
        s.setName(n);
        s.setMarks(m);
        s.displayDetails();
    }
}

class Student{
    private long rollNo;
    private String name;
    private double marks;

    public void setRollNo(long rollNo){
        this.rollNo=rollNo;
    }

    public void setName(String name){
        this.name=name;
    }

    public void setMarks(double marks){
        this.marks=marks;
    }

    public void displayDetails(){
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
    }
}
```

```
        System.out.println("Marks: " + marks);
    }
}
```

Input:

101
krishna
56.4

Output:

Roll No: 101
Name: krishna
Marks: 56.4

Q3. Student Details Using Default and Parameterized Constructors

Program:

```
import java.util.Scanner;

class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        long r=sc.nextLong();
        sc.nextLine();
        String n=sc.nextLine();
        double m=sc.nextDouble();

        Student s1=new Student();
        Student s2=new Student(r,n,m);

        s1.displayDetails();
        s2.displayDetails();
    }
}

class Student{
    long rollNo;
    String Name;
    Double marks;

    Student(){
        rollNo=101;
        Name="Rahul";
        marks=85.5;
    }
}
```

```

    }

    Student(long rollNo,String Name,double marks){
        this.rollNo=rollNo;
        this.Name=Name;
        this.marks=marks;
    }

    void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+Name);
        System.out.println("Marks: "+marks);
    }
}

```

Input:

101
Vin
98

Output:

Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 101
Name: Vin
Marks: 98.0

Q4. Net Salary Calculation Using Single Inheritance

Program:

```

import java.util.Scanner;

public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        Salary s1 = new Salary();
        int r = sc.nextInt();
        sc.nextLine();
        String n = sc.nextLine();
        double s = sc.nextDouble();

        s1.getEmployeeDetails(r, n, s);
        s1.calculateSalary();
    }
}

```

```

        s1.displayDetails();
    }
}

class Employee{
    int empId;
    String empName;
    double basicSalary;

    public void getEmployeeDetails(int empId, String empName, double basicSalary){
        this.empId = empId;
        this.empName = empName;
        this.basicSalary = basicSalary;
    }
}

class Salary extends Employee{
    double hra, da, netSalary;

    public void calculateSalary(){
        hra = basicSalary * 0.20;
        da = basicSalary * 0.10;
        netSalary = basicSalary + hra + da;
    }

    public void displayDetails(){
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("Net Salary: " + netSalary);
    }
}

```

Input:

1001

Vin

50000

Output:

Employee ID: 1001

Employee Name: Vin

Basic Salary: 50000.0

HRA: 10000.0

DA: 5000.0
Net Salary: 65000.0

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Program:

```
import java.util.Scanner;

class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        grossSalary g = new grossSalary();

        int a = sc.nextInt();
        sc.nextLine();
        String n = sc.nextLine();
        double b = sc.nextDouble();

        g.getEmployeeDetails(a, n);
        g.getBasicSalary(b);
        g.calculateSalary();
        g.displayDetails();
    }
}

class Employee{
    int empid;
    String empname;

    void getEmployeeDetails(int empid, String empname){
        this.empid = empid;
        this.empname = empname;
    }
}

class Salary extends Employee{
    double basicSalary;

    void getBasicSalary(double basicSalary){
        this.basicSalary = basicSalary;
    }
}

class grossSalary extends Salary{
```

```
double hra, da, grossSalary;

void calculateSalary(){
    hra = basicSalary * 0.20;
    da = basicSalary * 0.10;
    grossSalary = da + hra + basicSalary;
}

void displayDetails(){
    System.out.println("Employee ID: " + empid);
    System.out.println("Employee Name: " + empname);
    System.out.println("Basic Salary: " + basicSalary);
    System.out.println("HRA: " + hra);
    System.out.println("DA: " + da);
    System.out.println("Gross Salary: " + grossSalary);
}
}
```

Input:

5678

Vin

60000

Output:

Employee ID: 5678

Employee Name: Vin

Basic Salary: 60000.0

HRA: 12000.0

DA: 6000.0

Gross Salary: 78000.0